

Benchmarking sequence, embedding and structure-based methods for eukaryotic virus protein annotation

Lander De Coninck^{1*}, Uri Neri², Susie Grigson², Simon Roux²

¹KU Leuven, Microbiology, Immunology and Transplantation, Clinical and Epidemiological Virology, Herestraat 49, box 1040, Leuven, 3000, Belgium.

²Lawrence Berkeley National Laboratory, DOE Joint Genome Institute, Cyclotron Road 1, Berkeley, CA 94720, USA.

*Corresponding author(s). E-mail(s): lander.deconinck@kuleuven.be;

Contributing authors: uner@lbl.gov; sgrigson@lbl.gov; sroux@lbl.gov;

Abstract

The advent of protein language models and structural prediction promises new avenues for protein annotation through structure homology. Earlier work (eg. Phold) has shown protein structural homology annotation works well for prokaryotic viruses. To leverage these new techniques for protein annotation of eukaryotic viruses, we benchmarked four recent embedding- and structure-based methods in comparison with the gold standard of sequence-based annotation. We selected 10% of the entries in ICTV's VMR v40.2 that did not have bacteria or archaea as designated host (n=1,054), and subsequently retrieved all associated proteins from NCBI (n=11,360). Boltz-2 structure prediction of these proteins produced generally high-quality models (median pLDDT 71.3). Using the structures and protein sequences of the entries in the Big Fantastic Virus Database, we queried the VMR proteins with BLASTp, Diamond, and MMseqs2 (sequence); TEA and ProstT5-3Di (embedding); and Foldseek and Reseek (structure). Sequence homology-based methods showed a high overlap in identical hits, whereas the other methods had low intra- and inter-method agreement in hits between themselves and the sequence-based methods. Additionally, all methods improved the number of informative categories (ie. not "hypothetical protein") for the VMR proteins, with embedding and structure methods producing the largest increases. However, when evaluated against proteins with informative GenBank annotations as ground truth (n=2,791), embedding- and structure-based methods agreed less often with the "ground truth" than sequence-based methods (~67% and ~78% vs ~84%, respectively). Overall, sequence homology remains the most accurate approach for protein annotation of eukaryotic viruses at present, although for a minority of "hypothetical proteins" structure homology could help resolve functional annotation.

Introduction

The current increase in metagenomic sequencing experiments is providing researchers with a steady stream of new viral genomes. A large problem that remains for computational biologists

35 is the annotation of these genomes with functionally relevant labels for the predicted proteins. As
36 annotating all these genomes through experimental evidence is an unfeasible task, researchers
37 transfer annotations of annotated genomes to these novel genomes. The established gold
38 standard for performing this task has been the use of the BLAST suite or comparable sequence
39 alignment tools, such as Diamond and MMSeqs2¹⁻³. However, this approach reaches its limits
40 when newly discovered sequences diverge too much from experimentally validated annotated
41 genomes, a problem that is particularly acute in virology, where mutation rates are high and
42 sequence diversity is vast⁴.

43 Protein language models (pLMs) have emerged as a powerful paradigm for capturing biologi-
44 cally meaningful representations from amino acid sequences alone, trained on known protein
45 sequences using self-supervised objectives analogous to those used in natural language
46 processing^{5,6}. By learning the statistical regularities of protein sequences across billions of
47 examples, pLMs implicitly encode evolutionary, structural, and functional information in dense
48 embedding vectors. These representations have proven useful for a wide range of downstream
49 tasks, including secondary structure prediction, contact map prediction, solubility prediction,
50 and functional site annotation⁵⁻⁷. Models such as ESM-2⁷ and ProtTrans⁶ have become widely
51 used as general-purpose sequence encoders in the field.

52 A complementary and increasingly important line of work leverages protein three-dimensional
53 structure, which is generally more conserved than primary sequence over long evolution-
54 ary timescales, because the topology of a folded protein is directly tied to its biochemical
55 function⁸. The advent of highly accurate structure prediction tools such as AlphaFold2⁹ and,
56 more recently, Boltz-2¹⁰, has dramatically lowered the barrier to obtaining structural models.
57 Structural comparison tools, most notably Foldseek¹¹, have made it feasible to search large
58 structural databases at scale by encoding local backbone geometry into a compact structural
59 alphabet (3Di), enabling alignment-based searches analogous to BLAST but operating on struc-
60 ture rather than sequence. More recently, Reseek¹² was developed to further improve sensitivity
61 to remote structural homologs.

62 The convergence of pLMs and structural biology has opened a third avenue: using language
63 models to predict structural information without the computational cost of three-dimensional
64 structure prediction. ProstT5 is a notable example, fine-tuning a protein language model to
65 translate amino acid sequences into 3Di token sequences, effectively producing a structural
66 representation with a three order magnitude speedup compared to structure prediction¹³. A
67 related approach, The Embedded Alphabet (TEA), uses ESM-2 embeddings as input to generate
68 an alternative structural alphabet that the authors argue better represents helices and loops
69 compared to 3Di¹⁴. These compressed one-dimensional structural representations can then be
70 used as inputs to standard alignment tools, bridging the gap between pure sequence methods
71 and full structure comparison pipelines.

To address the annotation bottleneck for phage genomes specifically, Phold¹⁵ cleverly combines structural information with a large database of high-quality phage protein predictions. Phold either accepts user-provided structure predictions or uses ProstT5 to convert amino acid sequences into 3Di sequences, which are then aligned against annotated references using Foldseek. Benchmarking showed that Phold improved the annotation of a select set of bacteriophage proteins from approximately 35% to approximately 50% with both ProstT5 and protein structures¹⁵. A similar approach, Baktfold, was subsequently developed by the same group to increase annotation rates in bacterial genomes¹⁶.

In addition to bacteriophages, also less studied eukaryotic viruses (e.g. insect viruses, nematode viruses and plant viruses) might benefit from such an approach to bring a more functional understanding to metagenomic discoveries of these viruses. For example, the *Chrysoviridae*, a family infecting fungi, plants, and likely insects, whose four dsRNA genome segments encode four proteins. Two of these proteins — an RNA-dependent RNA polymerase and a capsid protein — have established functions, while the remaining two are of entirely unknown function¹⁷.

Therefore, we benchmarked seven approaches for functional annotation: BLASTp, Diamond and MMseqs2 as gold standard sequence alignment methods; ProstT5 3Di conversion with Foldseek alignment, and ESM-2 to The Embedded Alphabet (TEA) conversion with MMseqs2 alignment as pLM-derived structural alphabet approaches; and finally, full three-dimensional structure prediction of viral proteins with Boltz-2¹⁰ followed by structural alignment using either Foldseek¹¹ or Reseek¹². To evaluate these methods on a representative benchmark, we downloaded 11,360 proteins from 1,000 randomly selected, ICTV-ratified eukaryotic virus species, whose formal taxonomic recognition implies a better degree of annotation than randomly selected viral sequences from GenBank, while also being less comprehensively characterized than the curated entries in RefSeq. For all methods, we used the Big Fantastic Virus Database (BFVD), which contains 351,242 high-quality predicted viral protein structures, as a unified reference database¹⁸, ensuring that each annotation method was evaluated against an identical set of viral proteins and that observed differences in performance could be attributed to the annotation strategy rather than to differences in the underlying reference.

Results

Boltz-2 generates high-quality viral protein structures without parameter fine-tuning

Our test set of protein sequences consisted of 1,054 different viral entries taken at random from ICTV's VMR data (n=1,000 distinct species). These test dataset viruses covered all eukaryotic viral Kingdoms, with the majority of proteins originating from *Orthornavirae* (RNA viruses) and *Bamfordvirae* (dsDNA viruses) (see Figure 1A), despite the latter only representing 43 viruses (see Supplementary Figure 1).

Structure prediction of all 11,467 proteins from our test set with Boltz-2 resulted in high-quality predicted structures, with a median pLDDT score of 71.3 and 52.4% of all predicted structures having a pLDDT score above 70 (see Figure 1B). This shows that the structure prediction with default settings of Boltz-2 results in reliable structure predictions without the need for extensive fine-tuning of parameters, although extra viral sequences were used for the multiple sequence alignment (MSA) generation (see methods).

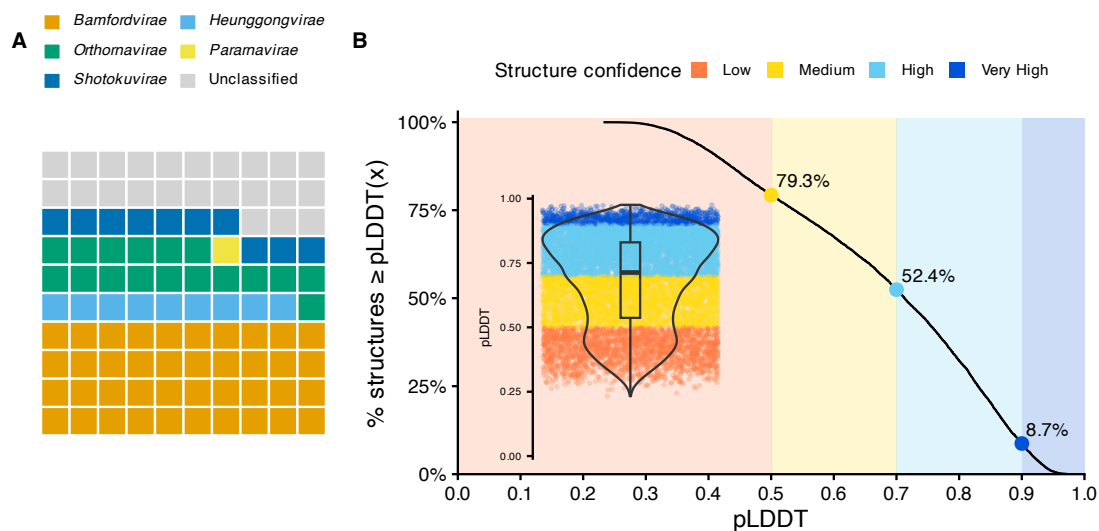


Figure 1: A) Test protein distribution from the different viral Kingdoms. Each tile roughly corresponds to 110 proteins. B) A cumulative distribution function plot showing the percentage of predicted structures from the test set with a pLDDT score above the pLDDT score on the x-axis. The inset shows a violin- and boxplot with all pLDDT scores from the predicted structures (n=11,467).

Sequence homology methods demonstrate superior sensitivity and reproducibility over structure-informed approaches in eukaryotic virus annotation

We started by evaluating how many of the proteins in our test produced at least one match against the BFVD for each method (see Figure 2A). In addition, we also filtered the results of each method on an E-value lower than 1e-03 to see how this affected the number of hits for each test protein. This decreased the number of test proteins with a hit on average with 25% for the structure methods. The methods relying on structure homology clearly had less matches compared to the sequence homology-based methods. This suggests that sequence-based methods are still more sensitive in detecting homology for eukaryotic viruses than structure-based methods, despite the promising results for bacteriophages¹⁵. As the quality of our structure predictions was rather high (see Figure 1B), this lower sensitivity is likely not

126 due to poor structure predictions. In contrast, TEA had, as only non-protein sequence-based
127 method, a comparable number of test proteins with a match, almost reaching the total number
128 of test proteins (n=11,360). However, analyzing the total number of hits at different E-value
129 thresholds shows that the TEA search resulted in a very high number of matches across all
130 E-value thresholds, signifying a high number of false positives (see Figure 2B). A similar pattern
131 was observed for the other structure-based methods, which tended to find too many hits at
132 high E-values. Reseek was less affected by this, due to a combination of already calculating
133 E-values more accurately^{12,19} and our use of a modified Reseek version (see below). On the
134 other hand, the three sequence-based methods had a similar number of test proteins with a
135 hit (BLASTp=11,359; MMseqs2=11,255 and Diamond=11,224), but don't show this inflation of
136 total hits at higher E-values (BLASTp had a slight increase towards larger E-values).

137 When comparing the BFVD matches yielded by all methods against each other (see Figure 2C-D
138 and Supplementary Figure 3), we observed that the sequence homology-based methods had a
139 high overlap in identical hits. This was true when comparing the best 25 hits for each test protein
140 (only the comparison of diamond and mmseqs2 against blastp had a low overlap because of
141 the higher amount of hits for blastp), the best 25 hits afterwards filtered on E-value, and only
142 the top hits (see Figure 2C-D and Supplementary Figure 3). In contrast, the embedding and
143 structure homology-based methods had low intra-method agreement in resulting hits (~25%
144 between the embedding methods and ~40-75% for the structure methods on the best 25 hits).
145 The percentage of overlapping hits did increase when only taking the top hits into account,
146 but stayed well below the percentage of overlap between the sequence-based methods. It is
147 worth noting that Reseek seemed to perform worse in this metric when looking at the filtered
148 results (see Supplementary Figure 3B), however this is due to the large reduction in number
149 of matches after filtering. This reduction is due to the more stringent E-value calculations in
150 Reseek compared to Foldseek^{12,19}. Notably, while ProstT5 was the worst performer across
151 the three datasets compared to the sequence methods, TEA performed better in the top hits
152 comparison and was on par with the structure-based methods. Overall, the embedding and
153 structure-based methods, that should be better to detect remote homologs, had low agreement
154 on shared hits among each other.

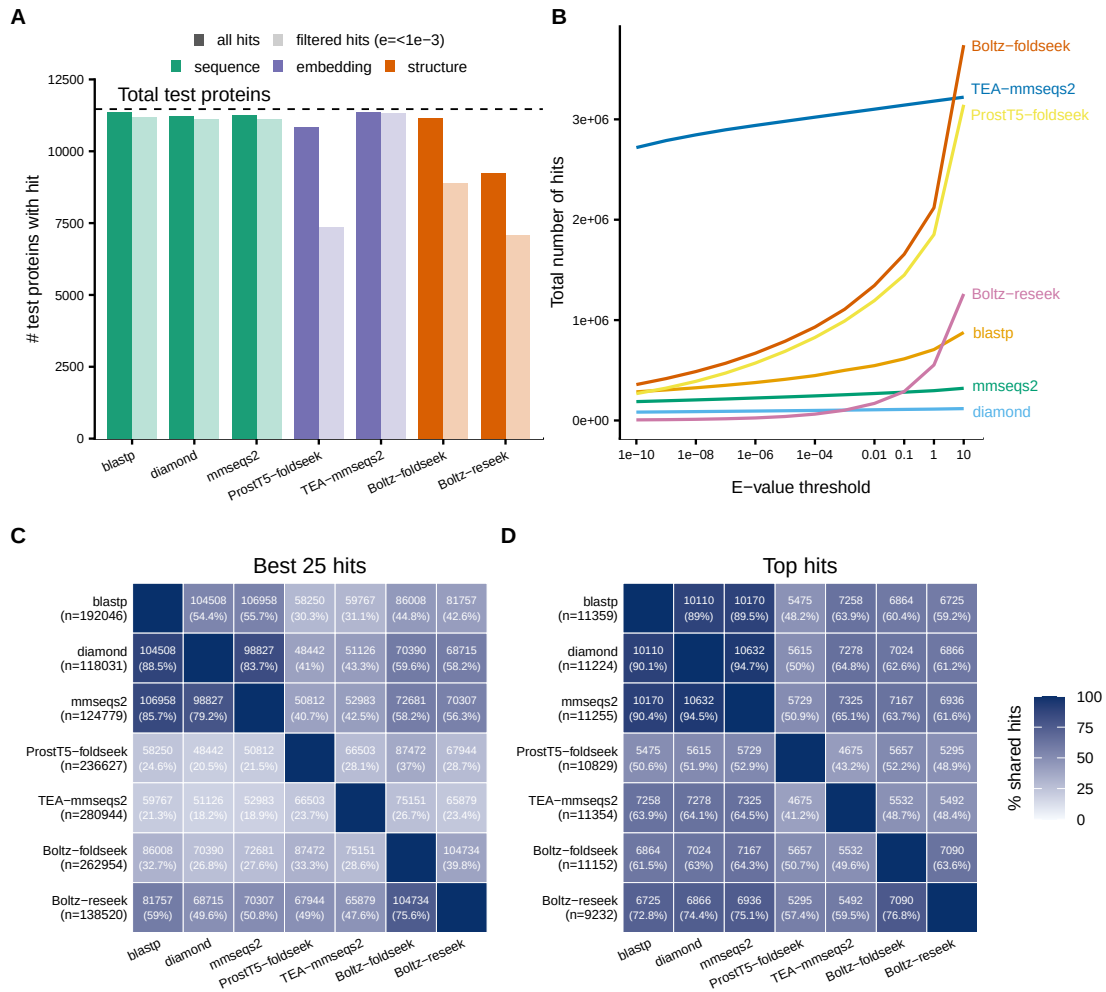


Figure 2: A) The number of proteins that got a hit against the BFVD shown per method (colored by category) before and after filtering the reported results on E-value lower than $1e-03$ (lighter color). B) The total number of hits across all test proteins in function of the E-value for each method. C-D) Pairwise comparisons of shared hits across methods for the same test proteins, shown for the best 25 hits (C), and the top hits regardless of E-value (D). Text inside the colored tiles indicate the number of shared hits for the method on the x-axis compared to the method on the y-axis, followed by the percentage of shared hits as a proportion of the total hits for the method on the y-axis.

Sequence-based methods retain better accuracy in functional category assignment while structural approaches slightly improve novel annotations

Using a weighted approach, we assigned functional categories to the test proteins for each method based on the hits with an E-value smaller than $1e-03$ (see methods). To evaluate the accuracy of the benchmarked methods, we considered the test proteins with an originally informative annotation ($n=2,938$) as the ground truth to compare our results to. All methods

161 improved the number of informative categories (see Figure 3A), with the sequence-based meth-
162 ods, together with TEA and Boltz-Foldseek, improving the number of informative assignments
163 the most. However, when comparing the functional annotations to the “ground truth”, all embed-
164 ding and structure-based methods underperformed compared to the sequence-based methods
165 (maximum ~67% and ~78% vs ~84% agreement with ground truth; see Figure 3B).

166 Although TEA made the largest improvements in assigning previous uninformative proteins to
167 informative categories, it also showed the greatest inconsistencies between assigned informative
168 categories that were already informative in the original test set (see Figure 3C). In addition,
169 it also has 387 queries that were changed from an informative function to a non-informative
170 one, which is almost double the amount compared to blastp which had the second highest
171 informative-to-uninformative reassignments (see Figure 3C and Table 1). The same applies
172 to the number of test proteins that originally had an informative category, but was assigned
173 a different category with TEA (n=522). Furthermore, due to the likely high number of false
174 positives in TEA hits (see Figure 2B), we concluded that the TEA uninformative-to-informative
175 assignments lacked reliability.

176 Of note, Boltz structure prediction combined with Foldseek did assign more informative cate-
177 gories than the sequence-based methods to originally non-informative proteins (n=1,571 vs
178 ~1,400; see Table 1). However, this method retrieved less test proteins with hits against the
179 BFVD.

- 180 • TODO: add that 337 could be assigned an informative function with Boltz-foldseek and not
181 with sequence-based methods
- 182 • most of them belong to structural categories (pvp / structural)
- 183 • TODO: make superpose figures with structures but filter first on true hypothetical

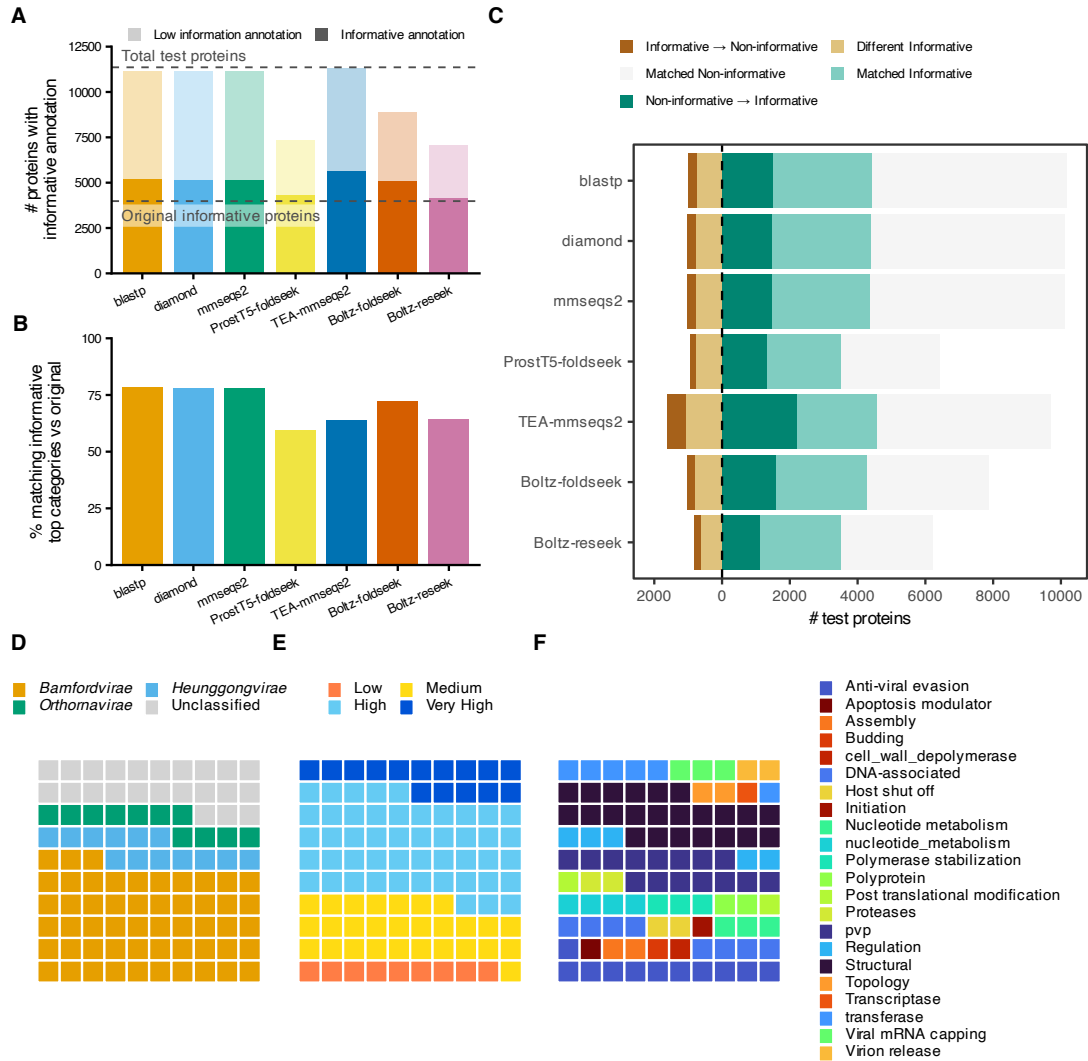


Figure 3: A) The number of proteins that received an informative functional annotation (i.e. not “Hypothetical Protein”, “unknown” or “Other Function”) after applying each method. Colored by method, light shades indicate non-informative proteins, dark shade indicates informative protein assignments. B) The percentage of queries that have a matching functional category for each method compared tot the informative original functional categories (regarded as the ground truth). C) Comparison of final assigned annotation categories compared to the original categories. Negative changes (ie. different informative categories or informative labels that changed to non-informative categories) are displayed to the left of the zero point, matches or category improvements (from non-informative to informative) are displayed on the right. D-E-F) Waffle plots showing the proportions of proteins that could be assigned an informative function with Boltz-foldseek and not with the sequence based methods, colored by D) taxonomy on Kingdom level, E) structure quality and F) main activity the assigned functional category takes part in (starts with uppercase; based on ViralZone) or the main category of the Empathi hierarchical classification system (starts with lowercase). Tiles are ordered from bottom to top, and from left to right. Each tile corresponds to roughly three proteins.

Table 1: Functional annotation outcomes across the seven benchmarked methods.

method	Queries assigned	Informative match	Uninformative match	Informative mismatch	Uninformative to informative	Informative to uninformative
blastp	11177	2946	5740	757	1477	257
diamond	11119	2934	5710	765	1462	248
mmseqs2	11135	2930	5731	768	1452	254
ProstT5-foldseek	7366	2198	2902	789	1313	164
TEA-mmseqs2	11318	2358	5132	1062	2208	558
Boltz-foldseek	8889	2703	3592	807	1578	209
Boltz-reseek	7065	2408	2710	635	1116	196

Computational benchmarking

All search methods were benchmarked for runtime and memory usage using Hyperfine²⁰, and `nvidia-smi` for GPU memory consumption. As expected, the sequence-based methods were much faster and less memory intensive compared to the embedding and structure-based methods (see Figure 4 and Supplementary Figure 4). Both Diamond and MMseqs2 finished the search of the >11,000 proteins against the BFVD in under 10 seconds, followed by BLASTp with a runtime of almost two minutes, a ~28x time increase compared to MMseqs2 which was the fastest workflow. The embedding methods took 52 and 93 times longer than MMseqs2 for TEA and the ProstT5 workflow, respectively. Sequence-based methods were generally compatible with consumer-grade hardware. In contrast, the TEA workflow using the full precision ESM2 model required substantially greater GPU VRAM, necessitating specialized hardware, whereas ProstT5 operated within the memory constraints of consumer-grade GPUs. The structural workflows, by contrast, required over 20,000-fold more runtime than the sequence-based tools (see Supplementary Figure 4) and demanded substantial computational resources, exceeding 400 GiB of system memory (CPU) and approximately 80 GiB of GPU memory (see Figure 4). Furthermore, the Reseek structural comparison incurred an additional two-hour runtime, whereas the Foldseek-based structural comparison completed in approximately two minutes.

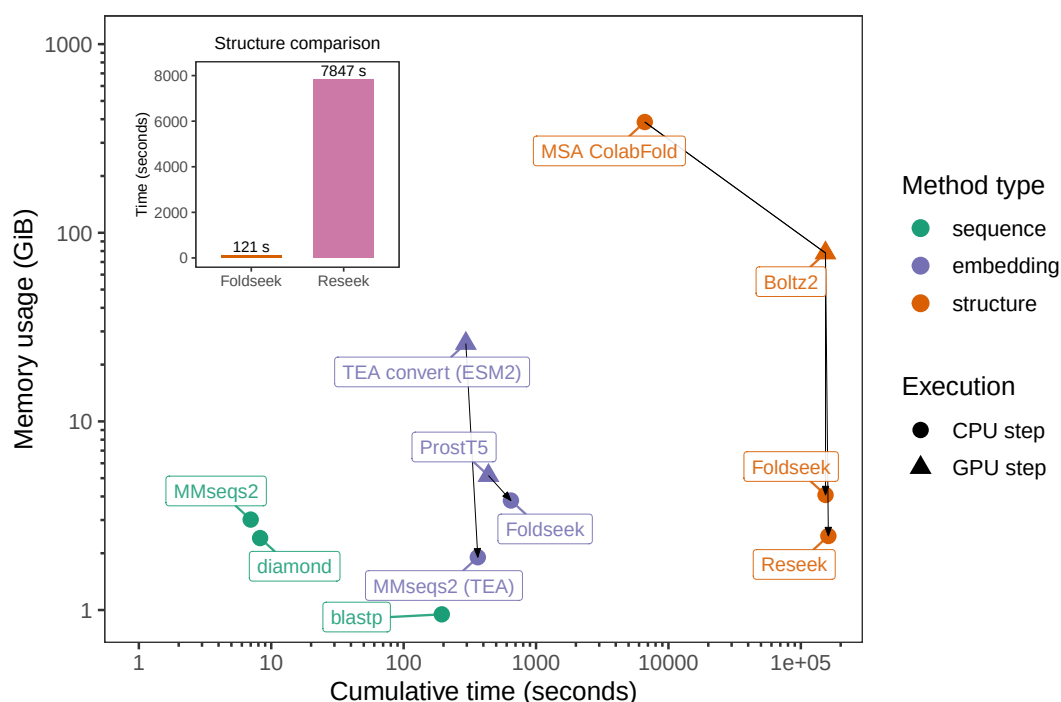


Figure 4: Log₁₀-transformed runtime in seconds of the different workflows plotted in function of the log₁₀-transformed memory usage (in GiB). For multistep workflows the indicated time is a cumulation of all steps (arrows indicate directionality of workflow steps). Workflows are colored by search method, CPU and GPU steps are indicated by circles and triangles respectively. The inset shows the difference in runtime between foldseek and reseek after MSA generation (ColabFold) and structure prediction (Boltz2).

Discussion

Although new methods like pLMs and structure homology searches have shown great promise in detecting remote homologs^{11,12,14}, including bacteriophage protein searches¹⁵, our benchmark suggests that these advantages do not yet translate straightforwardly to eukaryotic virus proteins. If we consider that structural homology searches are most useful to find homologs with a clearly defined protein structure, a possible explanation could be that most structural proteins (eg. capsid proteins) are already well annotated in sequence databases compared to more globular, complex proteins, and therefore the advantage of using structural homology searches in eukaryotic virus protein annotation might be limited.

- structure search is compatible with sequence search but retrieves little extra proteins (might be because of less virus proteins in training data)
- different conformational state predictions might hamper homolog detection (?)
- lack of virus representation in training data is a hurdle for the pLMs

- add note on latest papers showing decreased performance of pLMs for viruses (eg. mutation scanning)
- considerably higher runtime for little reward

Limitations: - polyproteins: cutting in pieces, not real function (collection of proteins with different functions) - classification (eg. nsp5), proteins same name different function -> finetune LLM on scientific papers - DB problem which is too limited (larger db might give better results)

Although protein language models and structure homology searches have shown great promise for detecting remote homologs^{11,12,14}, including in bacteriophage-focused annotation workflows¹⁵, our benchmark suggests that these advantages do not yet translate straightforwardly to eukaryotic virus proteins. Across our dataset, the sequence-based methods were consistently the most reproducible (high overlap of identical hits) and the most accurate when evaluated against the original classifications, whereas embedding- and structure-based methods showed lower agreement with the ground truth and substantially lower concordance with one another.

One plausible explanation is that the proteins for which structure-based searches are expected to excel (i.e. proteins with highly conserved folds despite limited sequence identity) may not be the dominant driver of annotation gaps in this particular setting. Many of the most abundant and best-characterized viral proteins, including major structural proteins such as capsid proteins, already have detectable sequence homology to entries in reference databases, so structure similarity is unlikely to provide large marginal gains for them. Conversely, proteins that remain annotated as “hypothetical” in eukaryotic virus genomes may include lineage-specific factors, short or low-complexity proteins, and rapidly evolving accessory proteins for which both sequence and structure homology signals can be weak, ambiguous, or absent. In this regime, structural comparison can still yield informative annotations for a subset of queries, but it does not consistently outperform sequence homology in either sensitivity or functional-category accuracy.

Importantly, our results do not imply that structure search is incompatible with sequence search; rather, the current value proposition appears to be complementary. A practical interpretation is that a sequence-first workflow remains the most reliable baseline, with structure-based approaches reserved for targeted follow-up of the subset of proteins that remain uninformative after stringent sequence searches. In our benchmark, structure-informed methods retrieved comparatively few additional proteins with high-confidence, interpretable hits once E-value filtering was applied, while incurring a very large computational cost. This cost-benefit imbalance is especially relevant for large-scale annotation projects, where throughput and reproducibility are essential.

Several methodological and biological factors could contribute to the lower apparent utility of embedding- and structure-based approaches for eukaryotic viruses in this benchmark. First,

252 both pLM-derived alphabets and structure-search pipelines ultimately depend on learned or
253 engineered representations that may be less well calibrated for viral proteins than for the broader
254 (mostly cellular) protein universe they were trained and tuned on. Limited viral representation in
255 training data could reduce the ability of pLMs to produce embeddings that cleanly separate
256 functionally distinct viral proteins, and could also affect downstream translation to structural
257 alphabets such as 3Di or TEA. In addition, the “many hits at high E-values” pattern observed for
258 some non-sequence methods suggests that statistical calibration and filtering thresholds may
259 require method- and domain-specific tuning to avoid inflation of weak or spurious matches.

260 Second, structural comparisons can be sensitive to conformational heterogeneity. Viral proteins
261 frequently operate in multi-protein assemblies and can adopt different conformational states
262 across the viral life cycle. Predicted structures represent a single (often lowest-energy) state,
263 which may differ from the biologically relevant state for binding or assembly. Even when overall
264 fold is correct, domain motions or alternative conformations could reduce similarity scores
265 for true homologs or increase similarity to unrelated proteins with superficially similar local
266 geometry, potentially lowering both sensitivity and the interpretability of structural hits.

267 Third, the reference database itself remains a key determinant of performance. Using BFVD
268 as a unified reference ensured a fair comparison between methods, but it also means that
269 any limitations in viral coverage, taxonomic representation, and annotation depth of BFVD will
270 propagate across all approaches. A larger and more diverse viral reference set—especially one
271 enriched for eukaryotic virus proteins and experimentally supported functional annotations—
272 could plausibly increase the payoff of structure-based searches. Similarly, improved viral-aware
273 scoring, E-value calibration, and domain-aware post-processing might yield higher-precision
274 structural or embedding-based annotations for the subset of proteins where sequence methods
275 fail.

276 These findings should be interpreted in light of several limitations of our benchmark. First, long
277 polyproteins were split into chunks to accommodate structure prediction constraints. While
278 necessary for computation, this can decouple functional context: individual fragments may
279 correspond to multiple mature proteins or partial domains, and their best-matching reference
280 hits may not reflect the final processed product or its function. Second, functional-category
281 assignment was derived from name-based classification rules applied to reference protein
282 names and to the original test annotations. This provides a scalable and consistent labeling
283 scheme, but it can conflate proteins that share a name while differing in function across
284 viral families (e.g. proteins labeled with the same “non-structural protein” identifier can have
285 different biochemical roles), and it can propagate ambiguities present in database annotations.
286 Incorporating richer context—such as genomic neighborhood, domain architecture, or literature-
287 informed curation (potentially via carefully constrained LLM-assisted extraction followed by
288 manual review)—could reduce these mismatches. Third, because all methods were evaluated
289 against a single unified database, our conclusions primarily reflect relative performance under a

290 controlled reference; performance may change with alternative databases, different taxonomic
291 compositions, or additional post-processing.

292 Overall, our benchmark supports a conservative conclusion: for eukaryotic virus protein anno-
293 tation, sequence homology remains the most accurate and reproducible primary strategy at
294 present. Embedding- and structure-based methods can still add value as complementary
295 tools for a minority of difficult proteins, but realizing their full promise will likely require viral-
296 aware training data, improved statistical calibration, and reference databases with broader viral
297 diversity and deeper functional validation.

298 **Material and Methods**

299 **Dataset selection**

300 **Test set**

301 To benchmark different methods for eukaryotic virus protein annotation, we selected 10% of
302 the entries in ICTV's Virus Metadata Resource (VMR) v40.2 that did not have bacteria or archaea
303 as designated host (n=1,054; 1,000 distinct species). We subsequently extracted their GenBank
304 accession numbers (n=1,402) and downloaded all associated proteins (n=11,360).

305 Protein sequences that were longer than 2,500 amino acids were split into multiple chunks of
306 maximum 2,500 amino acids so that structure prediction with Boltz-2¹⁰ would not run out of
307 GPU memory (see below). This resulted in a final test set of 11,467 protein sequences from
308 1,000 eukaryotic virus species.

309 **Database**

310 The BFVD¹⁸ was used as the reference database for both sequence and structure homology-
311 based annotation. The BFVD is a comprehensive database of high-quality predicted protein
312 structures from the viral sequence representatives of the UniRef30 clusters. UniRef30 is a
313 UniProt database that clusters protein sequences at 30% sequence identity and selects a
314 representative sequence for each cluster, these representatives thus have a UniProt accession.
315 Since a substantial amount of BFVD's included proteins are no longer available in UniProtKB as
316 they are not part of reference proteomes, we converted their UniProt accessions to UniParc IDs.

317 The BFVD database was further converted into the required formats the search tools used in
318 this benchmarking effort (see sequence and structure homology-based annotation).

319 **Functional classifier**

320 Functional categories for each BFVD entry were derived from protein names associated with
321 the corresponding UniParc IDs (could be from UniProtKB/TrEMBL, RefSeq and/or GenBank) for
322 eukaryotic viruses or by classifying them into PHROG-like²¹ categories with Empathi²² in the

323 case of bacteriophage proteins. The distinction between eukaryotic virus and bacteriophage
324 proteins was made based on the TaxID information from the UniParc protein ID.

325 Functional categories for eukaryotic virus proteins were designed based on previously estab-
326 lished enzymes by ViralZone (see <https://viralzone.expasy.org/10417>) and adding structural
327 proteins together with extra enzymatic categories. These extra categories were acquired by
328 prompting Perplexity AI in deep research mode with:

329 “Can you give me an exhaustive list of common protein functions in eukaryotic viruses? Use <https://viralzone.expasy.org> and <https://ictv.global/report/genome> as sources. Give me a machine readable file
330 with all functional categories.”
331

332 The resulting functional categories were further manually curated to ensure that they were
333 mutually exclusive. The final list of functional categories is provided in Table 2. A custom python
334 script was used to assign the established functional categories to the eukaryotic virus BFVD
335 proteins using Google’s Gemma4 31B-it LLM model. The LLM had access to the collection
336 of protein names and the viral taxonomy of each entry, and was asked to provide a category
337 classification based on this information. Assigned functional categories were checked against
338 the originally established functional categories and removed if they did not correspond.

339 The same functional classifier script was also used on the protein names from our test set
340 to assign functional categories to the eukaryotic virus proteins downloaded from the ICTV
341 exemplar species.

342 An overview of the LLM pipeline is given in Supplementary Figure 5

343 **Sequence homology-based annotation**

344 For sequence homology annotation, we used three state-of-the-art tools widely used in func-
345 tional annotation pipelines and sequence similarity searches. As gold standard, BLASTp
346 v2.16.0+¹ was used to align the test set amino acid sequences against the amino acid sequences
347 of the BFVD entries. In addition, we used `diamond blastp` (Diamond v2.1.16)²³ in sensitive
348 mode and `mmseqs easy-search` (MMseqs2 v18.8cc5c)³ with default settings; except for an
349 E-value cutoff of 10 instead of 1e-03 to allow comparison to the other methods.

350 **Protein embedding-based annotation**

351 **ProstT5**

352 ProstT5¹³ is a pLM, based on ProtT5-XL-U50⁶, finetuned for translating amino acid sequences
353 into structure representations with 3Di sequences. We used ProstT5 within foldseek
354 v10.941cd33¹¹ using default settings of `foldseek createdb` to generate 3Di sequences for our
355 test set and search them against the BFVD with `foldseek easy-search`.

356 TEA

357 In contrast to ProstT5, TEA¹⁴ uses the ESM2 pLM⁷ to generate embeddings for protein
358 sequences and converts these embeddings into TEA sequences using this novel structural
359 alphabet. Despite TEA and 3Di both consisting of 20 characters, TEA is different from 3Di
360 in terms of the numbers and distribution these TEA characters map to secondary structural
361 elements, promising a better representation of different structural elements across multiple
362 alphabet characters.

363 We converted both our test set and the BFVD into TEA sequences using the `tea_convert`
364 pipeline with default settings, and subsequently performed sequence similarity searches using
365 `mmseqs easy-search`³ with a custom substitution matrix (matcha substitution matrix provided
366 with TEA) and gap scoring settings (`--comp-bias-corr 0 --mask 0 --gap-open 18 --gap-`
367 `extend 3`) to search the test set against the BFVD.

368 Structure homology-based annotation

369 Structure prediction

370 For the structure prediction of our test set proteins, we relied on Boltz v2.2.1¹⁰ (default settings)
371 with MSAs generated by a local, adapted ColabFold installation²⁴ (see <https://github.com/bouras13/colabfoldv>). To increase the depth and diversity of the MSAs, we used the default
372 ColabFold sequence databases, supplemented with extra *Orthornavirae* sequences²⁵ and viral
373 sequences used in the structure prediction for the Phold database¹⁵.

375 Foldseek

376 Foldseek¹¹ converts predicted protein structures into 3Di sequences and subsequently performs
377 fast structure similarity searches using these 3Di sequences. Here, we used `foldseek easy-`
378 `search` (Foldseek v10.941cd33) with default settings to search the predicted structures of our
379 test set against the BFVD.

380 Reseek

381 Reseek¹² aligns structures in a similar way to Foldseek, but uses a different structural alphabet
382 consisting of $\sim 10^{11}$ states, promising better sensitivity for remote homologs.

383 We used an adapted version of Reseek v2.7 in sensitive mode that switches the loading of the
384 queries and database to get more accurate E-value calculations (see [this discussion](#)).

385 Functional category assignment

386 To functionally annotate our test proteins for each method, weights in form of the negative log
387 E-value (or a weight of 1000 in case of an E-value of 0) were given to each hit. These weights
388 were summed per functional category (assigned to each entry in the BFVD; see above) and the

category with the largest weight was chosen as the functional category for each test protein. Informative categories were prioritized over uninformative categories (ie. “Hypothetical Protein”, “unknown”, “Other function”) when the weight of the informative category was at least 50% the weight of the uninformative categories.

Computational benchmarking

All steps in our analysis were benchmarked for both time and memory usage using hyperfine v1.20.0²⁰. GPU memory consumption was measured by sampling usage every 200 ms with nvidia-smi v580.82.07, recording the peak value observed after each step.

All CPU analysis steps were performed on an HPC system using one IceLake node with 72 cores (2 Intel Xeon Platinum 8360Y CPUs@2.4 GHz; 256 GiB RAM), except for the MSA generation step which was performed on an IceLake big memory node with 72 cores (2 Intel Xeon Platinum 8360Y CPUs@2.4 GHz; 2048 GiB RAM). The GPU analyses were performed on one NVIDIA H100 SXM5 GPU (80 GiB HBM3), mounted on a node equipped with 2 AMD EPYC 9334 CPUs (768 GiB RAM).

Code availability

Code to reproduce this analysis is available on [GitHub](#).

Data availability

upload to zenodo

Acknowledgements

LDC acknowledges support from the Research Foundation - Flanders (grants 11L1325N and V451025N). The resources and services used in this work were provided by the VSC (Flemish Supercomputer Center), funded by the Research Foundation - Flanders (FWO) and the Flemish Government.

Declaration of LLM usage

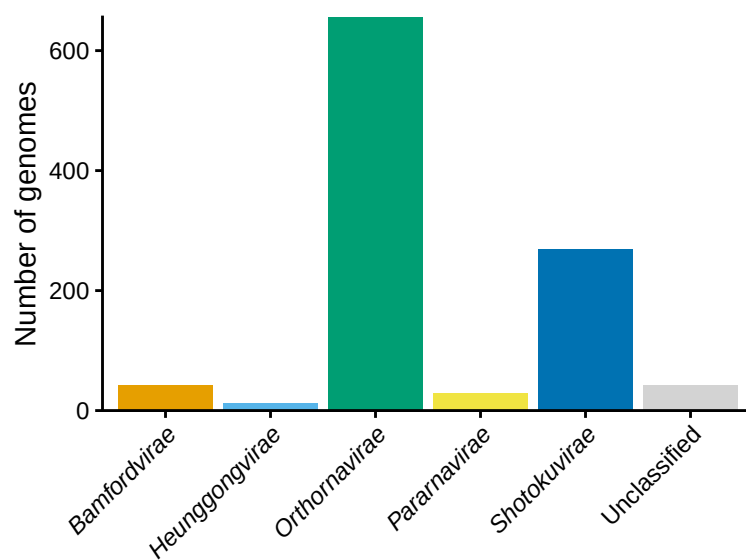
During the preparation of this manuscript, the authors made use of large language model (LLM)-based artificial intelligence tools to assist with several aspects of the writing and research process, including literature searches, restructuring and improving sentences, and generating code for data analysis. All AI-generated content and code was critically reviewed, verified, and edited by the authors, and full responsibility for the accuracy and integrity of the work rests with the authors alone.

419 References

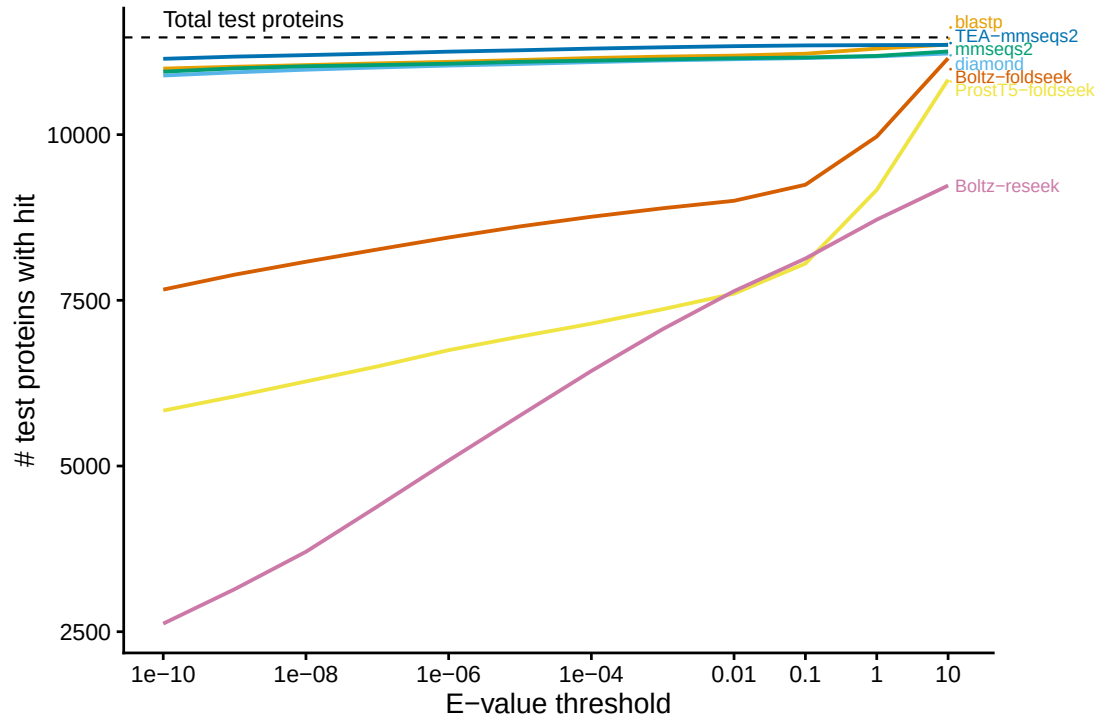
- 420 1. Altschul, S. F., Gish, W., Miller, W., Myers, E. W. & Lipman, D. J. [Basic local alignment search tool](#). **215**, 403–410 (1990).
- 421 2. Buchfink, B., Xie, C. & Huson, D. H. [Fast and sensitive protein alignment using DIAMOND](#). *Nature Methods* **12**, 59–60 (2014).
- 422 3. Steinegger, M. & Söding, J. [MMseqs2 enables sensitive protein sequence searching for the analysis of massive data sets](#). *Nature Biotechnology* **35**, 1026–1028 (2017).
- 423 4. Simmonds, P. *et al.* [Virus taxonomy in the age of metagenomics](#). *Nature Reviews Microbiology* **15**, 161–168 (2017).
- 424 5. Rives, A. *et al.* [Biological structure and function emerge from scaling unsupervised learning to 250 million protein sequences](#). *Proceedings of the National Academy of Sciences* **118**, e2016239118 (2021).
- 425 6. Elnaggar, A. *et al.* [ProtTrans: Toward Understanding the Language of Life Through Self-Supervised Learning](#). *IEEE Transactions on Pattern Analysis and Machine Intelligence* **44**, 7112–7127 (2022).
- 426 7. Lin, Z. *et al.* [Evolutionary-scale prediction of atomic-level protein structure with a language model](#). *Science* **379**, 1123–1130 (2023).
- 427 8. Illergård, K., Ardell, D. H. & Elofsson, A. [Structure is three to ten times more conserved than sequence—A study of structural response in protein cores](#). *Proteins: Structure, Function, and Bioinformatics* **77**, 499–508 (2009).
- 428 9. Jumper, J. *et al.* [Highly accurate protein structure prediction with AlphaFold](#). *Nature* **596**, 583–589 (2021).
- 429 10. Passaro, S. *et al.* Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction. 2025.06.14.659707 (2025) doi:[10.1101/2025.06.14.659707](#).
- 430 11. van Kempen, M. *et al.* [Fast and accurate protein structure search with Foldseek](#). *Nature Biotechnology* **42**, 243–246 (2024).
- 431 12. Edgar, R. C. [Protein structure alignment by Reseek improves sensitivity to remote homologs](#). *Bioinformatics* **40**, btae687 (2024).
- 432 13. Heinzinger, M. *et al.* [Bilingual language model for protein sequence and structure](#). *NAR Genomics and Bioinformatics* **6**, lqae150 (2024).
- 433 14. Pantolini, L. *et al.* Rewriting protein alphabets with language models. 2025.11.27.690975 (2026) doi:[10.1101/2025.11.27.690975](#).
- 434 15. Bouras, G. *et al.* [Protein structure-informed bacteriophage genome annotation with Phold](#). *Nucleic Acids Research* **54**, (2026).
- 435 16. Bouras, G. *et al.* Baktfold: Sensitive protein functional annotation across the microbial tree of life using structural information. 2026.03.31.715528 (2026) doi:[10.64898/2026.03.31.715528](#).

- 436 17. Kotta-Loizou, I. *et al.* [ICTV virus taxonomy profile: Chrysoviridae](#). *Journal of General Virology* **101**, 143–144 (2020).
- 437 18. Kim, R. S., Levy Karin, E., Mirdita, M., Chikhi, R. & Steinegger, M. [BFVD—a large repository of predicted viral protein structures](#). *Nucleic Acids Research* **53**, D340–D347 (2025).
- 438 19. Edgar, R. C. & Sahakyan, H. Protein structure alignment significance is often exaggerated. 2025.07.17.665375 (2025) doi:[10.1101/2025.07.17.665375](#).
- 439 20. Peter, D. Hyperfine. (2023).
- 440 21. Terzian, P. *et al.* [PHROG: Families of prokaryotic virus proteins clustered using remote homology](#). *NAR Genomics and Bioinformatics* **3**, lqab067 (2021).
- 441 22. Boulay, A., Leprince, A., Enault, F., Rousseau, E. & Galiez, C. [Empathi: Embedding-based phage protein annotation tool by hierarchical assignment](#). *Nature Communications* **16**, 9114 (2025).
- 442 23. Buchfink, B., Reuter, K. & Drost, H.-G. [Sensitive protein alignments at tree-of-life scale using DIAMOND](#). *Nature Methods* **18**, 366–368 (2021).
- 443 24. Mirdita, M. *et al.* [ColabFold: Making protein folding accessible to all](#). *Nature Methods* **19**, 679–682 (2022).
- 444 25. Mutz, P. *et al.* [The protein structurome of Orthornavirae and its dark matter](#). *mBio* **16**, e03200–24 (2024).

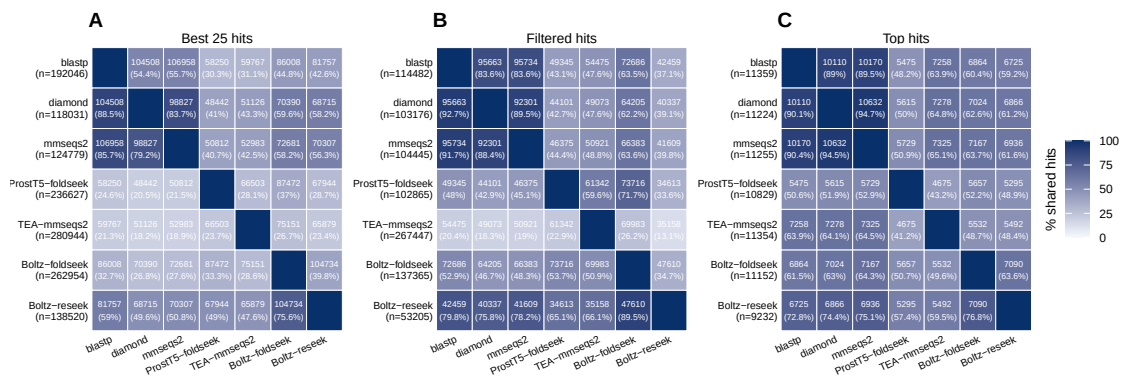
Supplementary Figures



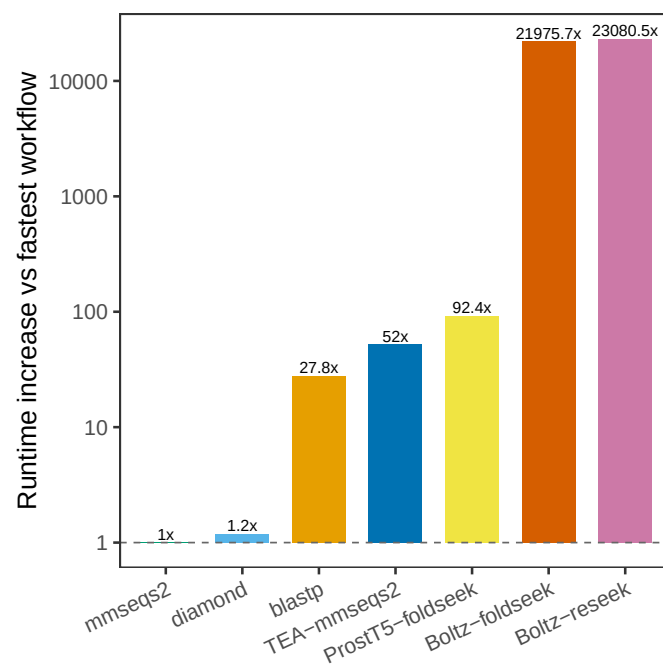
Supplementary Figure 1: The distribution of the number of genomes from the different viral Kingdoms in the test set.



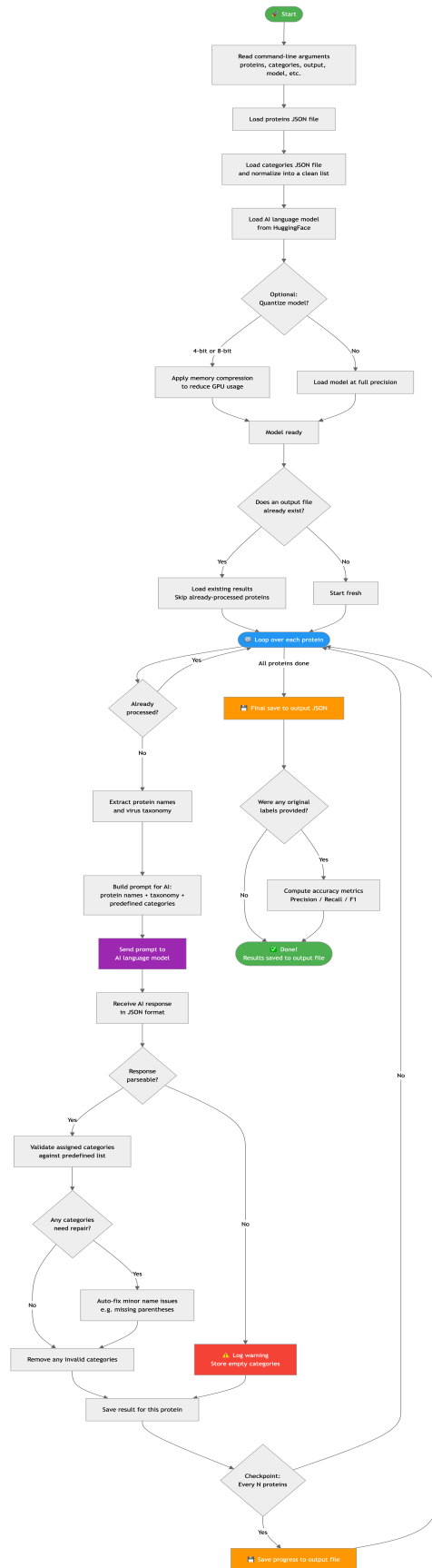
Supplementary Figure 2: The number of test proteins with at least 1 hit for each method across different E-value thresholds.



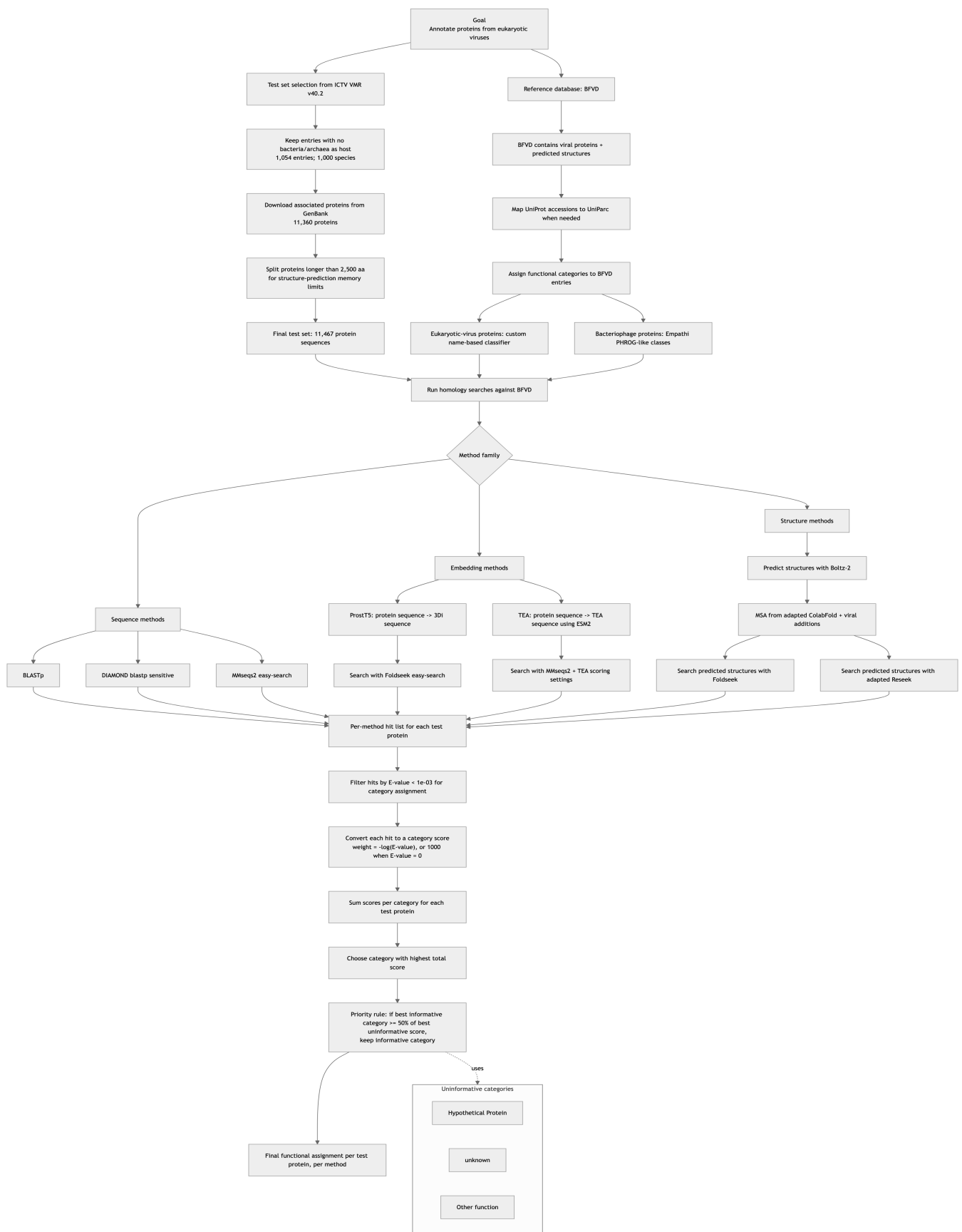
Supplementary Figure 3: Pairwise comparisons of shared hits between search methods for (A) the best 25 hits, (B) the best 25 hits after filtering on E-value ($E < 1e-03$), and (C) the top hits for each query.



Supplementary Figure 4: Workflow runtime increase compared to the fastest workflow (MMseqs2).



Supplementary Figure 5: Flowchart of the LLM classification pipeline for eukaryotic viruses.



Supplementary Figure 6: Flowchart of the annotation pipeline used in this study. Eukaryotic virus proteins were selected from ICTV VMR v40.2, filtered to exclude bacteria- and archaea-hosted entries, and expanded into a final test set after retrieving all associated protein sequences and splitting very long proteins for structure prediction. The BFVD reference database was prepared separately by assigning functional categories to its entries using a custom name-based classifier for eukaryotic virus proteins and PHROG-like categories for bacteriophage proteins by Empathi. The test proteins were then compared against BFVD using three sequence methods (BLASTp, DIAMOND, MMseqs2), two embedding-based methods (ProstT5-3Di and TEA), and two structure-based methods (Foldseek and Reseek after Boltz-2 structure prediction). Hits were filtered by E-value, converted into weighted category scores, summed per functional category, and the highest-scoring category was chosen for each test protein, with a rule that preserves informative categories when they are sufficiently supported over uninformative ones.

Supplementary Tables

Table 2: Final functional categories used for annotation, including their activity and process they are involved in.

Process	Activity	Enzyme/Protein
Entry	Fusion	Fusion glycoprotein
Exit	Budding	Phospholipase
Exit	Virion release	Neuraminidase
Exit	Ion-channel	Viroporin
Exit	Budding	Virion release facilitator
Exit	Virion release	Movement protein
Other to be classified	Host shut off	Trans-prenyltransferase
Other to be classified	Host shut off	Deoxyuridine 5'-triphosphate nucleotidohydrolase
Other to be classified	Host shut off	Ribonucleoside-diphosphate reductase
Other to be classified	Host shut off	FAD-linked sulfhydryl oxidase
Other to be classified	Host shut off	Nucleoside triphosphatase
Other to be classified	Host shut off	Alkaline nuclease
Other to be classified	Host shut off	ADP-ribose hydrolase
Other to be classified	Host shut off	dUTP nucleotidohydrolase
Other to be classified	Host shut off	Uracil-DNA glycosylase
Polyprotein	Polyprotein	Polyprotein
Protein	Proteases	Cys/Ser-proteases
Protein	Proteases	Asp-proteases
Protein	Proteases	Metallo-proteases
Protein	Proteases	Asparagine-proteases
Protein	Post translational modification	Beta-1,6-N-acetyl-glucosaminyl transferase
Protein	Post translational modification	Peptide N-myristoyl transferase
Protein	Post translational modification	thiol oxidase
Protein	Post translational modification	RNA-binding regulatory protein
Structural	Structural	Capsid protein
Structural	Structural	Nucleocapsid protein
Structural	Structural	Surface/envelope glycoprotein
Structural	Structural	Matrix protein
Structural	Structural	Tegument protein
Structural	Structural	Occlusion Body Protein
Unknown	Unknown	Hypothetical protein
Viral replication	Polymerase	DNA dependent DNA polymerase (DNA replicase)
Viral replication	Initiation	Primase
Viral replication	Initiation	Protein-primed initiation of DNA replication

Viral replication	Initiation	Protein-primed initiation of RNA replication (=VPG capping)
Viral replication	Initiation	Cress-type Endonuclease
Viral replication	Initiation	Ribozyme Endonuclease
Viral replication	Elongation	RNA-DNA duplex dependent endoribonuclease (RNase H)
Viral replication	Topology	DNA topoisomerase
Viral replication	Topology	DNA ligase
Viral replication	Topology	Helicase
Viral replication	Topology	Holliday junction resolvase
Viral replication	Topology	Integrase
Viral replication	Topology	Transposase (DDE recombinase)
Viral replication	Nucleotide metabolism	Thymidine kinase
Viral replication	Nucleotide metabolism	Thymidylate synthase
Viral replication	Nucleotide metabolism	Thymidylate kinase
Viral replication	Nucleotide metabolism	dUTP diphosphatase
Viral replication	Nucleotide metabolism	dihydrofolate reductase
Viral replication	Nucleotide metabolism	Ribonucleotide reductase
Viral replication	Regulation	E3 SUMO-protein ligase
Viral replication	Polymerase stabilization	Replication cofactor
Viral replication/transcription	Polymerase/Transcriptase	RNA dependent RNA polymerase (RNA replicase)
Viral replication/transcription	Polymerase/Transcriptase	RNA dependent DNA polymerase (Reverse transcriptase)
Viral transcription	Transcriptase	Proofreading exoribonuclease (ExoN)
Viral transcription	Viral mRNA capping	Cell-type capping
Viral transcription	Viral mRNA capping	Triphosphatase
Viral transcription	Viral mRNA capping	GTase
Viral transcription	Viral mRNA capping	N7-MTase
Viral transcription	Viral mRNA capping	2'O Mtase
Viral transcription	Viral mRNA capping	Alpha type capping
Viral transcription	Viral mRNA capping	Nido-type capping
Viral transcription	Viral mRNA capping	Mononega-type capping
Viral transcription	Viral mRNA capping	VPG capping
Viral transcription	Viral mRNA capping	Cap snatching
Viral transcription	Viral mRNA capping	Toti-type cap snatching
Viral transcription	Viral mRNA capping	FAD capping
Viral transcription	PolyAdenylation	Poly(A) polymerase
Viral transcription	PolyAdenylation	RNA polymerase stuttering
Viral transcription	Regulation	Transactivator protein
Viral transcription	Regulation	RNA export/splicing regulator
Viral-Host modulation	Anti-viral evasion	Polyuridine endonuclease

Viral-Host modulation	Anti-viral evasion	Viral ADP-Ribosylhydrolase
Viral-Host modulation	Anti-viral evasion	Asparagine deaminase
Viral-Host modulation	Anti-viral evasion	Ser/Thr kinase
Viral-Host modulation	Anti-viral evasion	Tyr kinase
Viral-Host modulation	Anti-viral evasion	Ubiquitin transferase=Ubiquitin-conjugating enzyme E2
Viral-Host modulation	Anti-viral evasion	ubiquitinyl hydrolase
Viral-Host modulation	Anti-viral evasion	Tyr/ser/thr phosphatase
Viral-Host modulation	Anti-viral evasion	2',3'-cGAMP endonuclease
Viral-Host modulation	Anti-viral evasion	3 beta-hydroxysteroid dehydrogenase/Delta 5-->4-isomerase
Viral-Host modulation	Anti-viral evasion	ADP-ribose hydrolase
Viral-Host modulation	Host shut off	mRNA-decapping
Viral-Host modulation	Apoptosis modulator	vCyclin
Viral-Host modulation	Apoptosis modulator	Apoptosis/cell-cycle modulator
Viral-Host modulation	Apoptosis modulator	vBcl-2
Viral-Host modulation	Anti-viral evasion	Interferon antagonist
Viral-Host modulation	Anti-viral evasion	Cytokine modulator
Virion assembly	Assembly	Assembly factor
